Standard Errors and Confidence Intervals for Direct, Indirect, and Total Effects in Continuous-Time Mediation Models

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This research was made possible by the Prevention and Methodology Training Program (PAMT) funded by a T32 training grant from the National Institute on Drug Abuse (NIDA).

Computations for this research were performed on the Pennsylvania State University's Institute for Computational and Data Sciences' Roar supercomputer.

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Abstract

Mediation modeling using longitudinal data is an exciting field that captures the interrelations in dynamic changes, such as mediated changes, over time. Even though discrete-time vector autoregressive (DT-VAR) approaches are commonly used to estimate indirect effects in longitudinal data, they have known limitations due to the dependency of inferential results on the time intervals between successive occasions and the assumption of regular spacing between measurements. To address these issues, continuous-time vector autoregressive (CT-VAR) models have been proposed as an alternative. Previous work in the area (e.g., Deboeck & Preacher, 2015; Ryan & Hamaker, 2021) has shown how the direct, indirect, and total effects, for a range of time-interval values, can be calculated using parameters estimated from CT-VAR models for causal inferential purposes. However, methods for calculating the uncertainty around the direct, indirect, and total effects in continuous-time mediation have yet to be explored. Drawing from the mediation model literature, we present and compare results from using the delta and Monte Carlo methods to calculate standard errors and confidence intervals for the direct, indirect, and total effects in continuous-time mediation for inferential purposes. Options to automate these inferential procedures and facilitate interpretations are available in the `cTMed` `R` package.

Keywords: mediation, continuous-time, delta method, Monte Carlo method, R package, cTMed

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Hypothetical Example 2: Quantifying Uncertainty in Direct, Indirect, and Total Mediation Effects in CT-VAR Models

In the previous section, we showed how to calculate point estimates for the direct, indirect, and total effect for a given time-interval from the CT-VAR drift matrix $\mathbf{\Phi}$. In particular, we know the direct, indirect, and total effects of marital conflict (η_X) on behavioral competence (η_Y) through maternal knowledge (η_M), for a time-interval of 3, are -0.145, -0.027, and -0.172, respectively. However, we still do not have a way to assess these effects from an inferential point of view. In this section, we extend the previous section by providing methods for quantifying the uncertainty in the direct, indirect, and total effects we have defined. In the following sections, we introduce the delta and the Monte Carlo methods. We chose these approaches because they are relatively easy to compute even for complex models. We also wish to contrast the performance of a method where symmetry is imposed versus and a method where asymmetry is accounted for. We discuss these methods in general and then apply them to the context of generating standard errors and confidence intervals as a way to quantify the uncertainty for the direct, indirect, and total effects.

The Delta Method

The delta method is a powerful statistical technique used to derive the asymptotic distribution of a random variable. It becomes particularly handy when dealing with functions of other random variables that are asymptotically Gaussian, that is, their distributions converge to a Gaussian distribution as sample size increases. As previously mentioned, it has been applied in psychological research to generate standard errors and confidence intervals for the indirect effect (Aroian, 1947; Goodman, 1960; Sobel, 1986, 1987). It has also been applied to other functions of parameters such as the standardized regression coefficients (Jones & Waller, 2015; Pesigan et al., 2023; Yuan & Chan, 2011).

Currently, popular SEM software packages such as `Mplus` (Muthén & Muthén, 2017) and `lavaan` (Rosseel, 2012) use the delta method as the standard approach to generating standard errors and confidence intervals for defined functions of SEM parameters. In `R`, the `Delta` and `DeltaGeneric` functions in the `betaDelta` (Pesigan et al., 2023) package can generate standard errors and confidence intervals for functions of parameters of fitted models as long as the estimates and the sampling variance-covariance matrix are provided.

We define the delta method following the multivariate central limit theorem. Let $\boldsymbol{\theta}$ be $\operatorname{vec}(\boldsymbol{\Phi})$, that is, the elements of the drift matrix $\boldsymbol{\Phi}$ in vector form sorted column-wise. Let $\hat{\boldsymbol{\theta}}$ be $\operatorname{vec}\left(\hat{\boldsymbol{\Phi}}\right)$, that is the estimates of the elements of the drift matrix. The function $\mathbf{g}$using $\hat{\boldsymbol{\theta}}$ as input can be expressed as:

$$\sqrt{n}\left(\mathbf{g}\left(\hat{\boldsymbol{\theta}}\right)-\mathbf{g}\left(\boldsymbol{\theta}\right)\right)\xrightarrow{\mathrm{D}}\mathcal{N}\left(0,\mathbf{J}\boldsymbol{\Gamma}\mathbf{J}'\right) \tag{1}$$

where $\mathbf{J}$ is the matrix of first-order derivatives of the function $\mathbf{g}$ with respect to the elements of $\boldsymbol{\theta}$ and $\boldsymbol{\Gamma}$ is the asymptotic variance-covariance matrix of $\hat{\boldsymbol{\theta}}$. From the former, we can derive the distribution of $\mathbf{g}\left(\hat{\boldsymbol{\theta}}\right)$ for a sample size n as follows

$$\mathbf{g}\left(\hat{\boldsymbol{\theta}}\right)\approx\mathcal{N}\left(\mathbf{g}\left(\boldsymbol{\theta}\right),n^{-1}\mathbf{J}\boldsymbol{\Gamma}\mathbf{J}'\right). \tag{2}$$

The uncertainty associated with the estimator $\mathbf{g}\left(\hat{\boldsymbol{\theta}}\right)$ is, therefore, given by $n^{-1}\mathbf{J}\boldsymbol{\Gamma}\mathbf{J}'$. When $\boldsymbol{\Gamma}$ is unknown, by substitution, we can use the estimated sampling variance-covariance matrix of $\hat{\boldsymbol{\theta}}$, that is, $\hat{\mathbb{V}}\left(\hat{\boldsymbol{\theta}}\right)$ for $n^{-1}\boldsymbol{\Gamma}$. Therefore, the sampling variance-covariance matrix of $\mathbf{g}\left(\hat{\boldsymbol{\theta}}\right)$ is given by

$$\mathbf{g}\left(\hat{\boldsymbol{\theta}}\right)\approx\mathcal{N}\left(\mathbf{g}\left(\boldsymbol{\theta}\right),\mathbf{J}\hat{\mathbb{V}}\left(\hat{\boldsymbol{\theta}}\right)\mathbf{J}'\right). \tag{3}$$

From this general approach, we can generate the sampling variance-covariance matrix of any function of the drift matrix $\boldsymbol{\Phi}$. The standard errors, that is, the square-root of the diagonal elements of the sampling variance-covariance matrix is used to generate

symmetric confidence intervals around the point estimates of any function of the drift matrix $\mathbf{\Phi}$. In current application, let the function $\mathbf{g}$ be a vector of the direct, indirect, and total effects of a specific independent variable on a dependent variable through one or more mediator variables. The `DeltaMed` function in the `cTMed` package applies the delta method to generate standard errors, test statistics, p-values and confidence intervals for these effects for a specific time-interval or a range of time-intervals.

The Monte Carlo Method

The Monte Carlo method was introduced in the mediation literature by MacKinnon et al. (2004) and formalized by Preacher and Selig (2012) as a resampling technique for generating confidence intervals for the indirect effect. It has been shown to be as good as nonparametric bootstrap (Pesigan & Cheung, 2023; Tofighi & Kelley, 2019, 2020; Tofighi & MacKinnon, 2015). As a resampling technique, it is similar to the bootstrap such that the sampling distribution of the function of parameters does not need to be normal. Unlike the bootstrap, however, it is more computationally efficient as it does not involve multiple model fitting. The `R` package `semmcci` provides a general approach to applying the Monte Carlo method for generating confidence intervals for a function of any SEM parameters. The `MCFunc` and `MCGeneric` functions in the `semmcci` (Pesigan & Cheung, 2023) package can generate standard errors and confidence intervals for functions of parameters of fitted models as long as the estimates and the sampling variance-covariance matrix are provided.

Based on the asymptotic properties of maximum likelihood estimators, we can assume that they are normally distributed around the population parameters.

$$\hat{\boldsymbol{\theta}} \sim \mathcal{N}\left(\boldsymbol{\theta}, \mathbb{V}\left(\hat{\boldsymbol{\theta}}\right)\right) \tag{4}$$

Using this distributional assumption, a sampling distribution of $\hat{\boldsymbol{\theta}}$ which we refer to as $\hat{\boldsymbol{\theta}}^*$ can be generated by replacing the population parameters with sample estimates, that is,

$$\hat{\boldsymbol{\theta}}^{*} \sim \mathcal{N}\left(\hat{\boldsymbol{\theta}}, \hat{\mathbb{V}}\left(\hat{\boldsymbol{\theta}}\right)\right). \tag{5}$$

A sampling distribution of $\mathbf{g}\left(\hat{\boldsymbol{\theta}}\right)$, which we refer to as $\mathbf{g}\left(\hat{\boldsymbol{\theta}}^{*}\right)$, can be generated by using the simulated estimates to calculate $\mathbf{g}$. The standard deviations of the simulated estimates are the standard errors. In resampling approaches, it is customary to use the percentiles of the generated distribution as confidence intervals instead of the standard errors. This approach allows the confidence limits to be asymmetric around the point estimates unlike the delta method which is always symmetric. Percentiles corresponding to $100\left(1-\alpha\right)\%$ are the confidence intervals.

From this general approach, we can generate the empirical sampling distribution of any function of the drift matrix $\boldsymbol{\Phi}$. There are constraints, however, on the randomly generated values of $\boldsymbol{\Phi}$. First, the diagonal values should be from zero to negative infinity. Second, the real part of all eigenvalues is less than zero. These two can be implemented using a replacement sampling approach, that is, discarding randomly generated values where the two constraints do not hold. Again, let the function $\mathbf{g}$ be a vector of the direct, indirect, and total effects of a specific independent variable on a dependent variable through one or more mediator variables. The `MCMed` function in the `cTMed` package applies the Monte Carlo method to generate the standard errors and confidence intervals for these effects for a specific time-interval or a range of time-intervals.

Confidence Intervals for the Direct, Indirect and Total Effects

Using the delta and Monte Carlo methods as implemented by the `cTMed` package, the standard errors and confidence intervals for the direct, indirect, and total effects of marital conflict (η_X) on behavioral competence (η_Y) through maternal knowledge (η_M), for time-intervals Δt of 1, 2, and 3 are presented in Table 1. We chose these time-intervals as they are the time-intervals available in the data set. Based on the results we highlight a few observations. The direct, indirect, and total effects changes across time-intervals. In

fact, the absolute value of the direct, indirect, and total effects increase as time-intervals increase implying stronger effects at larger time gaps between measurements. Looking at the confidence intervals, however, we can see that the direct and total effects are not statistically different from zero while the indirect effect is significant. We can see this as the confidence intervals for the indirect effect for all three time-intervals did not contain zero. From a substantive point of view, this shows that the association between marital conflict (η_X) and behavioral competence (η_Y) is mediated by maternal knowledge (η_M) given time-intervals of one, two, and three years.

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Table 1

Point, Standard Errors, and Interval Estimates of the Direct, Indirect, and Total Effects for Data Simulated from Grundy et al. (2007)

Effect	Estimate	Delta Method			Monte Carlo Method		
		Standard Error	2.5%	97.5%	Standard Error	2.5%	97.5%
			$\Delta t = 1$				
Direct	-0.077	0.023	-0.121	-0.033	0.023	-0.122	-0.033
Indirect	-0.006	0.003	-0.012	0.000	0.003	-0.012	0.000
Total	-0.083	0.022	-0.126	-0.040	0.022	-0.126	-0.039
			$\Delta t = 2$				
Direct	-0.121	0.036	-0.191	-0.051	0.036	-0.193	-0.051
Indirect	-0.016	0.009	-0.034	0.001	0.009	-0.034	0.001
Total	-0.137	0.035	-0.205	-0.070	0.035	-0.205	-0.069
			$\Delta t = 3$				
Direct	-0.145	0.044	-0.231	-0.059	0.044	-0.235	-0.061
Indirect	-0.027	0.014	-0.055	0.001	0.015	-0.055	0.003
Total	-0.172	0.042	-0.254	-0.090	0.042	-0.254	-0.090